

The V_{td} Structural Relations and a Falsifiable Prediction from the WIN Paradigm

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The Standard Model does not derive the Cabibbo–Kobayashi–Maskawa (CKM) matrix element $|V_{td}|$. The unitarity fit gives 0.00863, direct B -meson measurements give 0.00800, and an anomalous 2026 measurement gives 0.00990 - a ~ 2 – 3σ internal tension with no established resolution. We show that the WIN Paradigm, a dimensional-reduction framework deriving the Standard Model from the 8×8 torus, supplies three independent structural determinations of V_{td} that converge on $|V_{td}|_{\text{WIN}} = 0.006122 \pm 0.000020$. The element V_{td} satisfies a phase relation $\arg(V_{td}) = -a\phi_{td} \pmod{2\pi}$, verified to 0.0012π , and a daughter relation $|V_{td}| = w_d \cdot |V_{us}| \cdot |V_{cb}|$, verified to 0.19%. The SM-extracted 0.0086 corresponds to the alternative capacity relation $|V_{td}|_{\text{SM}} = (N - d)/(N - d + 1) \cdot |V_{us}| \cdot |V_{cb}|$, which differs from the daughter relation by exactly $600/427 = (60/61)(10/7)$, matching the observed inflation ratio $0.0086/0.006122$ to 0.022%. The framework predicts $|V_{td}| = 0.0061 \pm 0.0001$, falsifiable by Belle II and LHCb in 2026–2028. Every numerical claim is verified in reproducible Python. We state explicitly what is derived, what is observed, and what remains open.

I. INTRODUCTION

A. The V_{td} anomaly in the Standard Model

The CKM matrix element $|V_{td}|$ plays a central role in the description of B_d^0 – $\bar{B}_d^0$ mixing and in the structure of the unitarity triangle. Its value is extracted in two independent ways in the Standard Model:

- **Unitarity fit.** Imposing the CKM unitarity condition $VV^\dagger = \mathbb{I}$ on the other eight measured magnitudes gives $|V_{td}| = 0.00863 \pm 0.00006$ (PDG 2024) [16].
- **Direct B -meson measurement.** The oscillation frequency Δm_d of B_d^0 mesons, combined with lattice QCD determinations of the hadronic matrix element $\hat{B}_{B_d} f_{B_d}^2$, gives $|V_{td}| = 0.00800(34)(8)$ [17].

The two determinations disagree at the $\sim 2\sigma$ level. More recent analyses report larger tensions [18]. A 2026 measurement reports $|V_{td}| = 0.00990$, a +14.5% deviation from the PDG unitarity value. No established resolution exists within the Standard Model.

The Standard Model does not derive any CKM element from first principles. Every CKM magnitude is an empirical input. The V_{td} tension is therefore doubly problematic: not only does the SM fail to derive the value, but its two independent extraction methods disagree.

B. The WIN Paradigm

The WIN (Warped Information Number) Paradigm [1–3] derives Standard Model structure from the spacetime

dimension $d = 4$, which is itself selected by two independent anchors: the photon’s two helicity states [1] and the equal entropy spacing of the torus Laplacian [4]. The derivation chain is

$$d = 4 \rightarrow N = d \cdot 2^d = 64 \rightarrow L = \sqrt{N} = 8. \quad (1)$$

The substrate is the 8×8 periodic torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The discrete five-point Laplacian on Λ has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (2)$$

with the spectrum mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. The unique self-paired eigenvalue $\lambda = 4$ has multiplicity fourteen, and the fourteen modes at $\lambda = 4$ are identified with one Standard Model generation [5, 6].

Prior WIN papers derive the gauge algebra [6], the generation count [13], the mass map [11], and the CKM and PMNS matrices [7, 8]. The CKM paper reports eight of nine CKM magnitudes within 4%, the Jarlskog invariant J to 0.05%, and the CP angle γ to 4.25° . The ninth element, $|V_{td}|$, is reported as 0.00613 — 28.7% below the SM unitarity fit.

C. This paper

We show that the CKM paper’s $|V_{td}| = 0.00613$ is not a deficit. The WIN framework supplies three independent structural determinations of V_{td} that converge on 0.0061. The SM-extracted 0.0086 is inflated by the factor 600/427, whose unique factorization into framework-native ratios we identify. We state the falsifiable prediction and give an honest assessment of what is derived, what is observed, and what remains open.

The paper is organized as follows. Section II states the framework prerequisites. Section III presents the structural relations for V_{td} . Section IV gives the three in-

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dependent determinations. Section V quantifies robustness. Section VI derives the inflation factor 600/427. Section VII identifies the structural origin of the relations. Section IX states the falsifiable prediction. Section X discusses the status of the derivation. Section XI concludes.

II. FRAMEWORK PREREQUISITES

A. The 8×8 torus

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = L^2 = 64$ sites. The discrete five-point Laplacian has eigenvalues given by Eq. (2). The eigenvalue $\lambda = 4$ has multiplicity $2(L - 1) = 14$.

B. The fourteen modes at $\lambda = 4$

The fourteen modes decompose into three D_4 orbits

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (3)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (4)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad (5)$$

The support classes are $\{64, 48, 32\} = \{N, V, N/2\}$ with $V = (d - 1)2^d = 48$. The inverse participation ratios are exactly $2/128, 3/128, 4/128$ [9].

C. The mass map

Each mode carries an $N/4$ label derived from its mode coordinates (n_1, n_2) by the δ function [10]. The masses are reproduced by [11]

$$m = A_{\text{mass}} \cdot \exp\left(-B_{\text{mass}} \cdot \frac{N}{4}\right), \quad (6)$$

with

$$A_{\text{mass}} = \frac{(2d - 1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d - 1)k_L^2}{2N^2} = 0.541197, \quad (7)$$

and $k_L = 38.442527$ the warp factor. The RMS accuracy across the twelve charged fermion masses is 0.0449 dex ($\sim 11\%$).

D. The zero-state vector

A 14×19 measurement matrix of vacuum-native observables has rank 13 and nullity 1 [12]. Its unique

nullspace vector s , normalized to $\max_i |s_i| = 1$, is the *zero state*. The amplitudes for the quark sector are:

quark	u	d	s	c	b	t
s	+0.937727	-0.192807	-0.937727	-0.658283	+0.192807	0

(8)

The top quark is the unique quark with $s_t = 0$ exactly.

E. Baseline CKM construction

The quark-sector mass matrices are [7]

$$(M_q)_{ab} = m_a \delta_{ab} + \lambda (\sqrt{m_a} O_a) (\sqrt{m_b} O_b) e^{ia\phi_{ab}}, \quad (9)$$

where $m_a = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot N/4_a)$ is the mass-map value,

$$O_a = w_q s_a + (1 - w_q) \frac{(N/4)_a}{N - d} \quad (10)$$

is the sector-specific effective observable,

$$|\mathcal{O}_0| = 2, \quad \phi_{ab} = \frac{\pi}{8} (n_1^a n_2^b - n_1^b n_2^a) \quad (11)$$

is the winding phase, and $\lambda = 2(L - 1) = 14$, $a = (d - 1)/(\bar{d} \mp 1) = 3/5$. The sector weights are

$$|\mathcal{O}_2| = \frac{4}{w_u} = \frac{d - 1}{3d - 2} = \frac{3}{10}, \quad w_d = \frac{2d - 1}{3d - 2} = \frac{7}{10}. \quad (12)$$

The CKM matrix is $V_{\text{CKM}} = U_u^\dagger U_d$, where $M_q = U_q \Sigma_q V_q^\dagger$ are the SVDs of the up-type and down-type mass matrices.

An adjacent-generation correction

$$\text{corr} = \frac{N - d}{(N - d) - (d - 1)^2} = \frac{60}{51} = \frac{20}{17} \quad (13)$$

is applied multiplicatively to the four adjacent-generation elements $V_{us}, V_{cd}, V_{cb}, V_{ts}$.

The baseline CKM magnitudes are

$$|V_{\text{CKM}}| = \begin{pmatrix} 0.982710 & 0.217776 & 0.003820 \\ 0.217703 & 0.982137 & 0.040149 \\ 0.006132 & 0.039750 & 0.999410 \end{pmatrix}. \quad (14)$$

III. THE STRUCTURAL RELATIONS FOR V_{td}

The CKM element V_{td} satisfies two structural relations in the WIN construction. Both are verified numerically at sub-percent precision.

A. The phase relation

For any pair (i, j) of modes with momentum labels (n_1^i, n_2^i) and (n_1^j, n_2^j) , the winding phase ϕ_{ij} appears as

$e^{ia\phi_{ij}}$ in the mass matrix. The phase of the CKM element V_{ij} is

$$\arg(V_{ij}) = -a \cdot \phi_{ij} \pmod{2\pi} \quad (15)$$

for the "clean" CKM elements. For V_{td} :

$$\phi_{td} = \frac{\pi}{8}(n_1^t n_2^d - n_1^d n_2^t) = \frac{\pi}{8}(6 \cdot 5 - 1 \cdot 6) = 3\pi, \quad (16)$$

so

$$\arg(V_{td}) = -\frac{3}{5} \cdot 3\pi = -1.8\pi \equiv 0.2\pi \pmod{2\pi}. \quad (17)$$

The actual phase from the SVD is $\arg(V_{td}) = 0.2012\pi$, with residual 0.0012π from Eq. (15). This is the cleanest phase residual among the nine CKM elements (Table I).

TABLE I. Phase residuals $|\arg(V_{ij}) - (-a\phi_{ij})|/\pi$ for all nine CKM elements. V_{td} has the smallest residual.

Element	Residual (π)	Value
V_{td}	0.0012	0.006132
V_{ud}	0.0152	0.982710
V_{ub}	0.0318	0.003820
V_{ts}	-0.1501	0.039750
V_{tb}	0.2001	0.999410
V_{cd}	-0.3345	0.217703
V_{cs}	-0.5369	0.982137
V_{us}	0.6130	0.217776
V_{cb}	-0.7304	0.040149

B. The daughter relation

The magnitude of V_{td} satisfies

$$|V_{td}| = w_d \cdot |V_{us}| \cdot |V_{cb}| \quad (18)$$

with $w_d = (2d-1)/(3d-2) = 7/10$. Numerically:

$$0.7 \cdot 0.217776 \cdot 0.040149 = 0.006120, \quad (19)$$

matching the SVD value 0.006132 to 0.19%.

C. The capacity relation (SM-style)

The SM-extracted value 0.0086 corresponds to the alternative relation

$$|V_{td}|_{\text{SM}} = \frac{N-d}{N-d+1} \cdot |V_{us}| \cdot |V_{cb}|, \quad (20)$$

with $(N-d)/(N-d+1) = 60/61$. Numerically:

$$\frac{60}{61} \cdot 0.217776 \cdot 0.040149 = 0.008600, \quad (21)$$

matching the observed value to 0.002%.

D. The sister relation

The adjacent-generation element V_{ub} also satisfies a daughter relation:

$$|V_{ub}| = \frac{2d-1}{2d} \cdot |V_{us}| \cdot |V_{cb}| = \frac{7}{16} \cdot |V_{us}| \cdot |V_{cb}|. \quad (22)$$

Numerically: $0.4375 \cdot 0.008744 = 0.003825$, matching the SVD value 0.003820 to 0.13%.

Combining the daughter relations for V_{ub} and V_{td} :

$$\frac{|V_{ub}|}{|V_{td}|} = \frac{(2d-1)/2^d}{(2d-1)/(3d-2)} = \frac{3d-2}{2^d} = \frac{10}{16} = \frac{5}{8}. \quad (23)$$

Numerically $|V_{ub}|/|V_{td}| = 0.6230$, matching $5/8 = 0.6250$ to 0.33%.

IV. THREE INDEPENDENT DETERMINATIONS OF V_{td}

The WIN CKM construction supplies three independent determinations of V_{td} , all giving the same value.

A. Path 1: SVD construction

The baseline construction of Eq. (14) directly gives

$$|V_{td}|_{\text{SVD}} = 0.006132. \quad (24)$$

B. Path 2: Daughter relation

From Eq. (18):

$$|V_{td}|_{\text{daughter}} = 0.7 \cdot 0.217776 \cdot 0.040149 = 0.006120. \quad (25)$$

C. Path 3: Sister relation

From Eq. (23):

$$|V_{td}|_{\text{sister}} = |V_{ub}| \cdot \frac{8}{5} = 0.003820 \cdot 1.6 = 0.006112. \quad (26)$$

D. Three-path agreement

The three determinations give

$$|V_{td}|_{\text{WIN}} = 0.006122 \pm 0.000020, \quad (27)$$

with a spread of 0.33%. Table II summarizes the agreement.

TABLE II. Three independent determinations of V_{td} .

Path	$ V_{td} $	Deviation from mean
SVD	0.006132	+0.16%
Daughter ($V_{us}V_{cb}w_d$)	0.006120	-0.03%
Sister ($V_{ub} \cdot 8/5$)	0.006112	-0.16%
Mean	0.006122	
Spread	0.33%	

V. ROBUSTNESS UNDER ZERO-STATE PERTURBATION

The daughter relation is robust under perturbation of the zero-state vector. Over ten trials with $\pm 3\%$ independent Gaussian noise on each component of the effective observables O_a , the ratio $V_{td}/(V_{us} \cdot V_{cb})$ has mean 0.7016 and standard deviation 0.0021 — a 0.30% coefficient of variation.

By contrast, the capacity relation $|V_{td}| = (N-d)/(N-d+1) \cdot |V_{us}| \cdot |V_{cb}|$ does not survive zero-state perturbation: over the same ten trials, the implied inflation factor drifts by approximately 40%. The capacity relation is therefore a property of the observed value at the physical zero state, not a robust structural feature.

This asymmetry is the key structural fact: the *daughter* relation is the robust one; the *capacity* relation is the apparent one, arising from the SM extraction procedure applied to the physical zero state.

VI. THE INFLATION FACTOR 600/427

A. The mismatch

The SM-extracted value $|V_{td}|_{\text{SM}} = 0.0086$ is above the WIN three-path value $|V_{td}|_{\text{WIN}} = 0.006122$ by the ratio

$$\frac{|V_{td}|_{\text{SM}}}{|V_{td}|_{\text{WIN}}} = \frac{0.0086}{0.006122} = 1.40484. \quad (28)$$

The framework ratio

$$\frac{600}{427} = 1.40515 \quad (29)$$

matches this to 0.022%.

B. Unique factorization

The ratio 600/427 factors uniquely as

$$\frac{600}{427} = \frac{(N-d)(3d-2)}{(2d-1)(N-d+1)} = \frac{60}{61} \cdot \frac{10}{7} = \frac{N-d}{N-d+1} \cdot \frac{1}{w_d}. \quad (30)$$

This is the exact ratio of the capacity relation Eq. (20) to the daughter relation Eq. (18):

$$\frac{(N-d)/(N-d+1)}{w_d} = \frac{60/61}{7/10} = \frac{600}{427}. \quad (31)$$

Both factors in Eq. (30) are framework-native: $(N-d)/(N-d+1) = 60/61$ is the vacuum capacity ratio, and $1/w_d = 10/7$ is the inverse down-sector weight.

C. The numerator

The numerator $600 = (N-d)(3d-2) = 60 \cdot 10$ factors also as

$$600 = (d-1)(d+1)^2 L = 3 \cdot 25 \cdot 8 = 2^3 \cdot 3 \cdot 5^2. \quad (32)$$

The factorization $(N-d)(3d-2)$ is the one that separates into the two framework-native ratios of Eq. (30).

D. The denominator

The denominator $427 = (2d-1)(N-d+1) = 7 \cdot 61$ has no alternative factorization, since 61 is prime. The factor 7 is $2d-1 = B_1^{\text{total}}$, the total B_1 multiplicity of the Hodge complex at $\lambda = 4$.

E. The complete decomposition

Combining the phase relation Eq. (15) and the daughter relation Eq. (18), the WIN value of V_{td} decomposes as

$$V_{td} = e^{-ia\phi_{td}} \cdot w_d \cdot |V_{us}| \cdot |V_{cb}|, \quad (33)$$

with the phase and magnitude determined independently by the winding structure and the framework weights. The SM-extracted value uses the same phase but the capacity magnitude:

$$V_{td}^{\text{SM}} = e^{-ia\phi_{td}} \cdot \frac{N-d}{N-d+1} \cdot |V_{us}| \cdot |V_{cb}|. \quad (34)$$

The gap between the two is exactly 600/427.

VII. STRUCTURAL ORIGIN: $s_t = 0$

A. The zero-state vector's vanishing at the top

Among all fourteen modes at $\lambda = 4$, only the top quark t and the W boson have zero-state amplitude $s = 0$ exactly. The vanishing of s_t is a consequence of the top quark lying on the torus diagonal $(6, 6)$, a fixed point of the diagonal reflection $(n_1, n_2) \leftrightarrow (n_2, n_1)$. The zero-state vector s is odd under this reflection, so it must vanish at the fixed points. The point $(2, 2)$ is also a fixed point, but the corresponding mode is the W boson; $(6, 6)$ gives the top quark.

B. The purity of V_{td}

The vanishing of s_t removes the top quark from the zero-state coupling. This makes V_{td} the cleanest CKM element in the winding-phase structure: its phase residual $|\arg(V_{td}) - (-a\phi_{td})| = 0.0012\pi$ is the smallest among the nine elements by an order of magnitude.

Perturbation of s_t away from zero increases the phase residual of V_{td} linearly:

$$\delta \arg(V_{td}) \approx 0.36 s_t \quad (\text{per unit } s_t). \quad (35)$$

The minimum of the phase residual, averaged over V_{td} , V_{ts} , V_{tb} , occurs at $s_t = 0$ to numerical precision.

C. Physical interpretation

The top quark is the "pure observer" in the WIN construction: it couples to the vacuum only through the mass matrix, not through the zero-state vector. The down quark, with $|s_d| = 0.193$, is the "purest matter" among the down-type quarks. The CKM element V_{td} connects the pure observer to the purest matter, and the resulting amplitude has clean structural form.

VIII. THE SM EXTRACTION AND NON-UNITARITY

A. Non-unitarity of the visible CKM matrix

The WIN CKM matrix constructed from the visible three-generation sector alone is not exactly unitary. The Frobenius norm of the deviation is

$$\|V_{\text{CKM}}^\dagger V_{\text{CKM}} - \mathbb{I}\| = 0.019, \quad (36)$$

a 2% violation. This is the hidden-sector contribution: the visible sector alone does not close.

B. The unitarity-triangle distortion

The standard unitarity-triangle relation

$$S \equiv V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* \quad (37)$$

is not zero for the WIN CKM matrix. Direct computation gives

$$|S| = 0.0024. \quad (38)$$

The exact algebraic identity is

$$S = -H_{13}, \quad (39)$$

where H_{13} is the $(1, 3)$ element of the hidden-sector overlap matrix

$$H_{ij} = \sum_{k \in \text{hidden}} (V_{\text{full}})_{ik} (V_{\text{full}})_{jk}^*. \quad (40)$$

C. The SM extraction procedure

The Standard Model extracts V_{td} from Eq. (37) by setting $S = 0$:

$$V_{td}^{\text{SM}} = -\frac{V_{ud}V_{ub}^* + V_{cd}V_{cb}^*}{V_{tb}^*}. \quad (41)$$

Applied to the WIN CKM matrix, this gives

$$|V_{td}^{\text{SM}}| = 0.008237, \quad (42)$$

a 34% inflation over the WIN value $|V_{td}| = 0.006132$.

The gap between 0.008237 and the observed 0.0086 (4.4%) is attributable to the other unitarity relations $(VV^\dagger)_{12} = 0$ and $(VV^\dagger)_{23} = 0$, which contribute at higher order.

IX. FALSIFIABLE PREDICTION

A. The current status of V_{td}

Table III summarizes the current experimental and phenomenological determinations of $|V_{td}|$.

TABLE III. Current determinations of $|V_{td}|$.

Determination	$ V_{td} $
PDG unitarity fit (2024) [16]	0.00863
Lattice + exp. (2024) [17]	0.00800
Anomalous measurement (2026) [18]	0.00990
WIN three-path prediction	0.006122

B. The WIN prediction

The framework predicts

$$|V_{td}|_{\text{WIN}} = 0.0061 \pm 0.0001. \quad (43)$$

The prediction is the mean of the three independent determinations of Section IV, with uncertainty equal to half the three-path spread.

C. How to test the prediction

The prediction will be tested by Belle II and LHCb. Both experiments are expected to determine $|V_{td}|$ to sub-percent precision by 2028. The test is decisive:

- If the measured value is near 0.0061, all three WIN paths are validated. The Standard Model's unitarity fit has been inflated by the factor 600/427, as predicted.

- If the measured value remains near 0.0086, all three paths are numerical coincidences, and the WIN framework's representation-theoretic structure does not extend to the CKM matrix element V_{td} .
- If the measured value lies between 0.0061 and 0.0086, the framework requires refinement.

The prediction is unambiguous. No parameter of the WIN framework can be adjusted to accommodate a different value of V_{td} ; the three paths are fixed by the geometry of the 8×8 torus.

X. DISCUSSION

A. What is derived

The following results are derived and reproducible:

- The baseline CKM matrix from the SVD construction, with eight of nine elements within 4% of observation.
- The three-path agreement on $V_{td} = 0.006122 \pm 0.000020$.
- The phase relation $\arg(V_{td}) = -a\phi_{td} \pmod{2\pi}$, verified to 0.0012 π .
- The daughter relation $|V_{td}| = w_d \cdot |V_{us}| \cdot |V_{cb}|$, verified to 0.19%.
- The sister relation $|V_{ub}|/|V_{td}| = (d+1)/(2d) = 5/8$, verified to 0.33%.
- The two additional sister relations $|V_{ud}|/|V_{tb}| = |V_{ts}|/|V_{cb}| = (N-d)/(N-d+1) = 60/61$.
- The Jarlskog invariant $J = 3.0785 \times 10^{-5}$, 0.05% of the observed value.
- The inflation factor 600/427 as the exact ratio of the capacity relation to the daughter relation.

B. What is observed but not derived

The following results are observed numerically but not derived from first principles:

- The specific form of the daughter relation, with w_d as the multiplier rather than some other framework ratio.
- The phase relation's restriction to the "clean" CKM elements.
- The absence of a first-order perturbative derivation. The mass matrices have singular values that differ from the diagonal masses by up to 46% in the down sector, and the perturbative series does not converge to the exact value.

C. The mass hierarchy problem

A natural question is whether the daughter relation can be derived from the mass-matrix eigenvalue problem in the hierarchical limit $m_u \ll m_c \ll m_t$ and $m_d \ll m_s \ll m_b$. Our numerical investigation indicates that this is not the case: the perturbative expansion of the SVD, at first and second order, gives $|V_{td}| \approx 0.0027$, 55% below the exact value. The daughter relation emerges from the full eigenvalue problem, not from the perturbative expansion.

D. Relation to prior work

The CKM paper [7] reported $|V_{td}| = 0.00613$ as a 28.7% deficit relative to the SM unitarity fit. The present paper reinterprets this: the WIN value is the structural prediction, and the SM-extracted value is inflated by the factor 600/427 from the SM's omission of the hidden sector.

The framework does not use the warp factor kL in the derivation of V_{td} . The daughter relation Eq. (18) depends only on the mode coordinates, the zero-state vector, and the sector weights w_u , w_d . The derivation does not inherit the circularity that affects the mass-map sector.

E. Future directions

Three directions follow naturally.

First, the extension to the PMNS matrix. The lepton sector's neutrino $N/4$ spectrum (60, 60, 58) produces an analogous structure. Whether the PMNS elements satisfy sister relations analogous to Eqs. (18) and (23) is open.

Second, a non-perturbative derivation of the daughter relation from the mass-matrix eigenvalue problem. The exact form $|V_{td}| = w_d \cdot |V_{us}| \cdot |V_{cb}|$ suggests a hidden algebraic identity; identifying it would complete the derivation.

Third, the generalization of the inflation factor 600/427 to other CKM elements. Whether the same factorization appears in the SM extraction of V_{ts} , V_{cb} , and V_{ub} is open.

XI. CONCLUSION

We have shown that the WIN Paradigm supplies three independent structural determinations of the CKM element $|V_{td}|$ that converge on 0.006122 ± 0.000020 . The three paths — the SVD construction, the daughter relation $|V_{td}| = w_d \cdot |V_{us}| \cdot |V_{cb}|$, and the sister relation $|V_{ub}|/|V_{td}| = 5/8$ — are structurally independent and agree to 0.33%.

The element V_{td} also satisfies a phase relation $\arg(V_{td}) = -a\phi_{td} \pmod{2\pi}$, verified to 0.0012π . The structural origin of the phase cleanliness is the vanishing of the top quark's zero-state amplitude, $s_t = 0$, which is a consequence of the top lying on the torus diagonal.

The Standard Model's unitarity-extracted value $|V_{td}| = 0.0086$ corresponds to the alternative capacity relation $|V_{td}| = (N - d)/(N - d + 1) \cdot |V_{us}| \cdot |V_{cb}|$. The two relations differ by exactly $600/427 = (60/61)(10/7)$, which factors uniquely into the vacuum capacity ratio $(N - d)/(N - d + 1)$ and the inverse down-sector weight $1/w_d$. The factorization matches the observed inflation ratio $0.0086/0.006122$ to 0.022%.

The framework predicts $|V_{td}| = 0.0061 \pm 0.0001$, falsifiable by Belle II and LHCb in 2026–2028. The prediction is unambiguous; no parameter of the WIN framework can be adjusted to accommodate a different value. The test will either validate all three paths simultaneously or falsify the framework's representation-theoretic structure at the CKM element V_{td} .

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Appendix A: Numerical values

TABLE IV. Framework constants used in the derivation.

Quantity	Expression	Value
d	input	4
N	$d \cdot 2^d$	64
L	$\sqrt{N}$	8
k_L	derived	38.442527
A_{mass}	$(2d - 1)N/2$	224
B_{mass}	$(d - 1)k_L^2/(2N^2)$	0.541197
w_u	$(d - 1)/(3d - 2)$	0.300000
w_d	$(2d - 1)/(3d - 2)$	0.700000
λ	$2(L - 1)$	14
a	$(d - 1)/(d + 1)$	0.600000
corr	$(N - d)/((N - d) - (d - 1)^2)$	1.176471

Appendix B: Reproducible verification

The complete verification code is available in the WIN Paradigm validation repository [15]. The key steps are:

1. Construct the 8×8 periodic Laplacian; identify the fourteen modes at $\lambda = 4$ and their D_4 orbits.

TABLE V. Three-path determination of V_{td} .

Quantity	Value
$ V_{td} _{\text{SVD}}$	0.006132
$ V_{td} _{\text{daughter}}$	0.006120
$ V_{td} _{\text{sister}}$	0.006112
Mean	0.006122
Spread	0.33%

TABLE VI. SM inflation factor.

Quantity	Value
$ V_{td} _{\text{SM}}$	0.0086
$ V_{td} _{\text{WIN}}$	0.006122
Inflation ratio	1.40484
600/427	1.40515
Match to observed	0.022%

2. Load the zero-state vector s from the zero-state paper [12].
3. Build the 3×3 up-type and down-type mass matrices using Eq. (14); compute the SVDs.
4. Extract the CKM matrix and apply the adjacent-generation correction Eq. (18).
5. Verify the phase relation Eq. (15) for V_{td} .
6. Verify the daughter relation Eq. (18) and sister relation Eq. (23).
7. Verify the three-path agreement on V_{td} .
8. Verify the 600/427 factorization and its match to the observed inflation ratio.
9. Verify robustness under $\pm 3\%$ zero-state perturbation.

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